

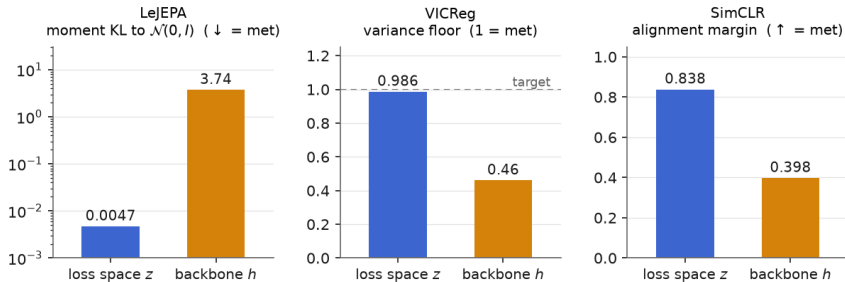
Two-Space Faithfulness in Projector-Based Self-Supervised Learning

The objective acts on $Z = p(H)$; downstream use keeps H .

1. the gap, and the bridge that relates the two spaces
2. a dual-space regularizer that controls what the bridge needs
3. how much augmentation variation H should retain — and what it buys

Each method's own promise: at z and h

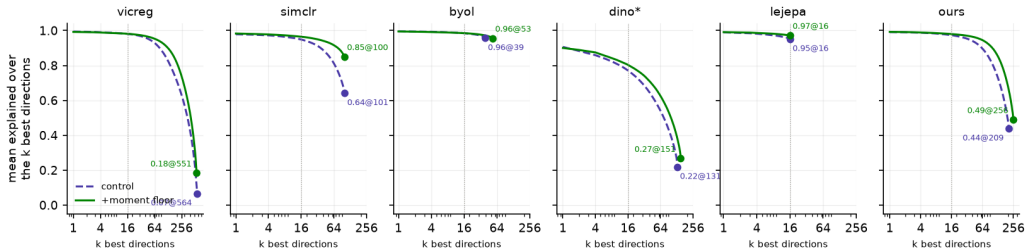
§1 *Properties guaranteed in the projector space need not hold in H .*



The bridge: how much of Z 's organization is linearly available from H ?

§1 Linear accessibility provides a way to relate the two spaces, but requires sufficient rank and dimensionality in both.

C3b — the accessibility profile (endpoint dot = bare R^2_{acc} at the method's own r_z ; dotted line = $k(16)$; IN-100)



Mean explained variance over the k best-explained whitened z directions vs. k (on held out data); each curve runs to that method's own target rank at z , its endpoint dot is R^2 . Violet dashed = control, green = + the h -side regularization.

The objective: one regularizer, applied in both spaces

§2 We introduce a dual-space regularizer with a spectral barrier against collapse. Applied to both H and Z , it directly controls the capacity required by the accessibility bridge.

$$\mathcal{L} = \underbrace{\mathcal{L}_{\text{pair}}(z_1, \dots, z_V)}_{\text{organizes } z} + \lambda_z \mathcal{K}(\hat{\mu}_z, \hat{\Sigma}_z) + \lambda_h \mathcal{K}(\hat{\mu}_h, \hat{\Sigma}_h),$$

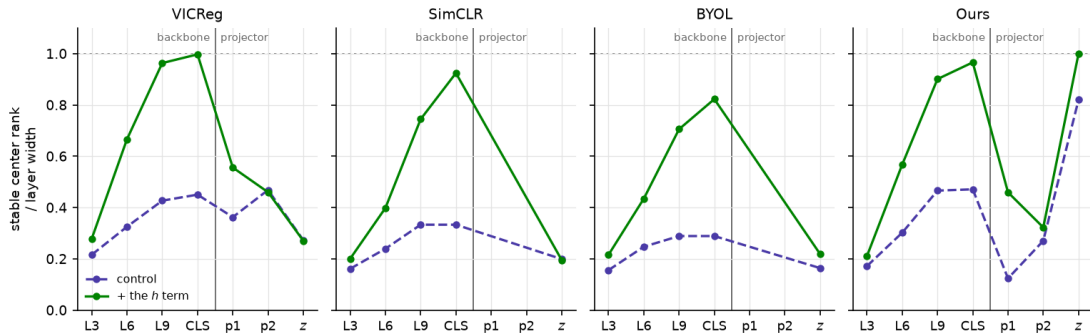
$$\mathcal{K}(\mu, \Sigma) = \frac{1}{2} (\|\mu\|^2 + \text{tr } \Sigma - \log \det \Sigma - d).$$

A **center** is one image's V augmented views, embedded and averaged. $\hat{\mu}_s, \hat{\Sigma}_s$ are the mean and covariance of those centers across a batch, in the backbone ($s = h$) or after the projector ($s = z$); d is that space's dimension.

- **The barrier.** $-\log \det \Sigma$ diverges as any eigenvalue of Σ approaches 0; $\text{tr } \Sigma$ penalizes the other side and $\|\mu\|^2$ pulls the mean to the origin.
- **Why centers.** $\text{Cov}(\text{centers}) = \text{Cov}(\text{exact centers}) + \frac{1}{V} \text{Cov}(\text{views of one image})$ — averaging first shrinks the augmentation part by $1/V$, so the term protects across-image spread rather than augmentation spread.
- **Both ends.** The same term at z and at h .

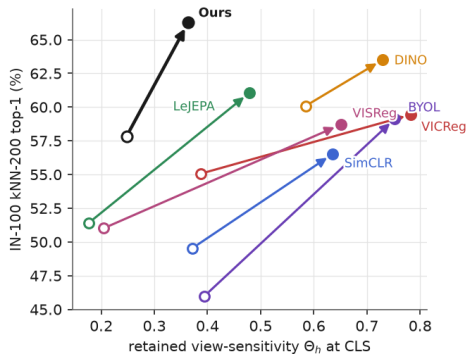
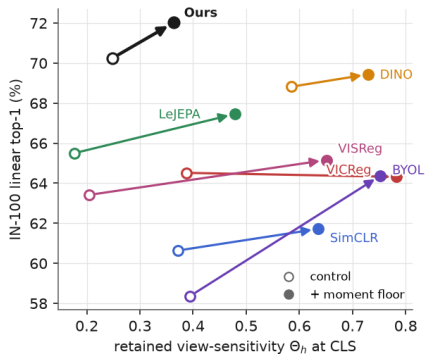
It protects stable center capacity in the retained backbone

§2 ... it controls the capacity required by the accessibility bridge.



Added to existing methods, it moves each one the same way

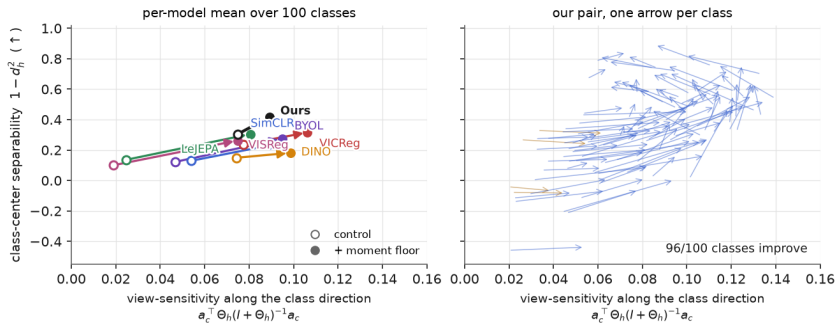
§3 *The same regularizer also changes how much augmentation variation is retained in H .*



One arrow per method, control $\rightarrow$ treated, in (retained view-sensitivity Θ_h at CLS, probe accuracy): linear top-1 left, kNN-200 right.

Where the gain comes from

§3 ... *neither too little nor too much is desirable.*



One arrow per model, control $\rightarrow$ treated (left: mean over the 100 classes; right: our pair, one arrow per class). **y-axis** = $1 - \text{MSE}$ (test) of a linear probe on view-averaged features. **Right** = what a *single* view costs relative to that average, along the class's own readout direction.

Competitive as a standalone objective (ImageNet-1k, one protocol)

§3 ... while achieving competitive performance as a standalone SSL objective.

Method	Backbone	Ep.	Linear	kNN
LeJEPA	ViT-S/16	100	64.1	47.1
DINO	ViT-S/16	100	70.0	64.7
iBOT	ViT-S/16	100	70.9	65.7
Ours	ViT-S/16	100	66.3	54.5
VISReg	ViT-B/16	100	70.3	—
Ours	ViT-B/16	100	71.2	56.0
Ours	ViT-L/16	100	—	—

Summary

1. Self-supervised learning objectives are typically enforced after a projector, while downstream tasks use the backbone representation H . Properties guaranteed in the projector space therefore need not hold in H . Linear accessibility provides a way to relate the two spaces, but requires sufficient rank and dimensionality in both.
2. Motivated by feature-learning results on the rank and dimensionality of learned representations, we introduce a dual-space regularizer with a spectral barrier against collapse. Applied to both H and Z , it directly controls the capacity required by the accessibility bridge.
3. The same regularizer also changes how much augmentation variation is retained in H . We show that this variation is meaningful relative to image-to-image variation, and that neither too little nor too much is desirable. Our method typically moves this ratio in a favorable direction while achieving competitive performance as a standalone SSL objective.